Define the Problem Statements

Field	Details
📅 **Date**	31 January 2025
🆔 **Team ID**	LTVIP2025TMID45577
📌 **Project Name**	Pollen's Profiling: Automated Classification of Pollen Grains
🎯 **Max Marks**	2 Marks

📝 Customer Problem Statement Template

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

🔗 **References**

- 📂 Dataset: https://www.kaggle.com/datasets/andrewmvd/brazilian-pollen-dataset
- 🧠 CNN (Sequential API): https://keras.io/api/models/sequential/
- 🌐 Flask Web Framework: https://flask.palletsprojects.com
- 🖼️ Image Augmentation (Keras): https://keras.io/api/preprocessing/image/

🔍 Customer Problem Statement

Manual classification of pollen grains is slow, error-prone, and dependent on trained specialists. This makes large-scale studies inefficient in the fields of environmental monitoring, allergy diagnostics, and agriculture.

Stakeholders such as ecologists, healthcare professionals, and agronomists require an automated solution to classify pollen samples quickly and accurately

from microscopic images.

💡 Proposed Solution

- Design and train a **Convolutional Neural Network (CNN)** model to classify pollen types from images.
- Use the **Brazilian Savannah Pollen Dataset** containing 790 images across 23 classes.

  📎 Kaggle Dataset
- Perform preprocessing: normalization, resizing to 128×128, label encoding.
- Train the model with:
  - Image augmentation
  - Class weighting
  - EarlyStopping callbacks
- Build a **Flask-based web application**:
  - Interface to upload pollen images.
  - Real-time classification and result display.

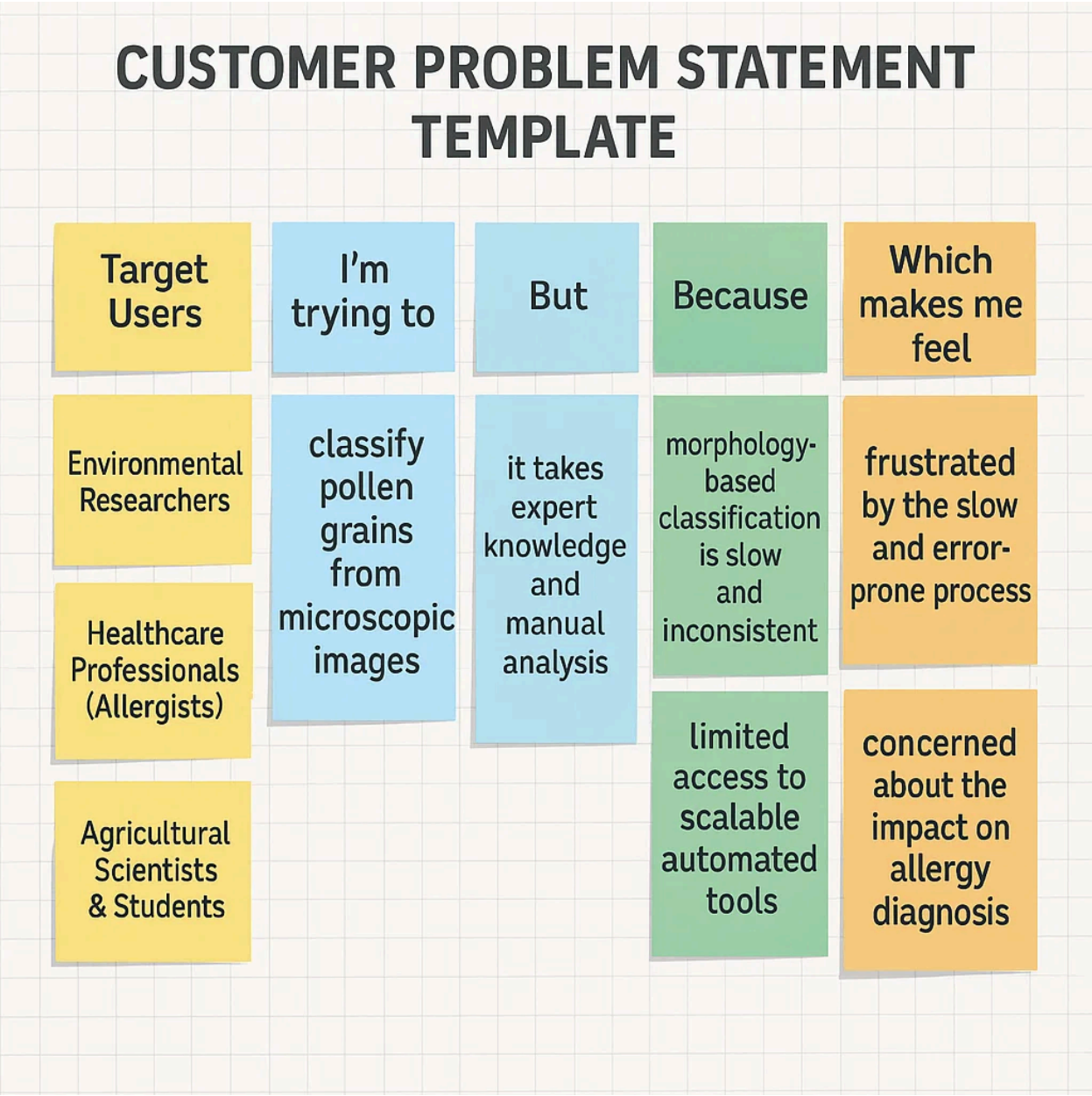
👥 Target Users

- 🌿 Environmental Researchers
- 🩺 Healthcare Professionals (Allergists)
- 🌾 Agricultural Scientists
- 🎓 Students & Academic Institutions

✅ Expected Outcome

- A deep learning model trained to identify 23 pollen grain classes.

  📎 CNN Guide: https://keras.io/guides/sequential_model/
- A Flask web application offering real-time predictions.

📎 Deployment: https://flask.palletsprojects.com/en/2.3.x/deploying/

- Time-saving and scalable solution for ecological, agricultural, and clinical use cases.